

COURSE DESCRIPTION			
Program	Instrumentation Engineering		
Course Title	Basics of Instrumentation and Transducers		
Course Code	2081	Semester	II
Type of Course	Theory	Contact Hours	60
Teaching Scheme (L: T:P:R)	4:0:0:0	Credits	4
CIE Marks	40	ESE Marks	60

Introduction

This course aims to give students a basic understanding of the fundamentals of Instrumentation Engineering and different types of transducers used in instrumentation systems.

Course Objectives

1	To introduce the basic concepts and principles of measurement and instrumentation.
2	To understand bridge circuits and display systems used in instrumentation.
3	To provide fundamental knowledge of transducers used for measuring physical quantities.
4	To learn different types of transducers and their basic applications.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Fundamentals of Electronic Instruments, passive and active components	1041	Elementary Concepts of Electronics	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Explain the basic concepts and principles of measurement and instrumentation.	Understand
CO2	Describe the working principles of basic bridge circuits and display systems used in instrumentation.	Understand
CO3	Explain the principles of Resistive transducers in a measurement system	Understand
CO4	Describe different types of transducers used for measuring physical quantities.	Understand

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	-	-	-	-	-	-	-	-	-	-
CO2	2	-	-	-	-	-	-	-	-	-	-
CO3	2	-	-	-	-	-	-	-	-	-	-
CO4	2	-	-	-	-	-	-	-	-	-	-
Total	8	-	-	-	-	-	-	-	-	-	-
Correlation Level	2	-	-	-	-	-	-	-	-	-	-

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
4	0	0	0	6	60	4	40	60	100	0	0	0	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus – Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Introduction to Measurement & Instrumentation	Definition of Measurement - Methods of measurement – Direct & Indirect, Absolute & Relative. Classification of Instruments – Deflection and Null type, Analog and digital instruments, Mechanical, Electrical and Electronic instruments. Block diagram of the Instrumentation system. Performance characteristics - Static (Accuracy, Precision, Resolution, Error, Sensitivity, Drift, Threshold, Range, Repeatability, Linearity, Hysteresis), Dynamic (Speed of response, Fidelity, Lag, Dynamic error) Types of Errors- Gross, Systematic and Random errors. Moving Coil and Moving Iron instruments. PMMC Galvanometer	17	0
Module II	Bridges and Display Devices	DC Bridge – Wheatstone's bridge (circuit and working only) AC Bridges- Maxwell's bridge, Hay's bridge, Schering bridge, Wien bridge (circuit and working only) Display devices – CRO, DSO, LED, LCD	11	0
Module III	Introduction to Transducers and Resistive Transducers	Classification of Transducers – on the basis of transduction form used, Passive and Active, Analog and Digital, Primary and Secondary, Mechanical and Electrical. Principle of Resistive transducer Linear Potentiometer - Flex transducer - LDR- Strain gauge - Types of strain gauges – Bonded, Unbonded, Semiconductor strain gauge. Weight measurement using Strain gauge bridge circuit.	15	0
Module IV	Different types of transducers	LVDT - Eddy current transducer - Capacitive transducer - Level measurement using capacitive transducer - Piezoelectric transducer - Hall effect transducer - Ultrasonic transducer - Level measurement using Ultrasonic transducer - Photoelectric transducers - Photodiode, phototransistor, Solar cell	17	0

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Basic definitions	Define the terms Instrumentation, Measurement and Instrument.	L	CO1	Understand	1
Measurement methods	Describe the various methods of measurement – Direct & indirect, absolute & relative.	L	CO1	Understand	2
Classification of Instruments	Identify the different classes of Instruments	L	CO1	Remember	1
Classification of Instruments	Describe the classes of instruments such as deflecting type and null type.	L	CO1	Understand	1
Classification of Instruments	Compare analog and digital instruments	L	CO1	Understand	1
Classification of Instruments	Explain electrical, mechanical and electronic instruments.	L	CO1	Understand	1
Block diagram of instrumentation system	Describe the functional elements of an instrument with block diagram	L	CO1	Understand	1
Performance characteristics of an instrument	Define static characteristics such as Accuracy, Precision, Resolution, Error, Sensitivity, Drift, Threshold, Range, Repeatability, Linearity and Hysteresis	L	CO1	Understand	2
Performance characteristics of an instrument	Define various dynamic characteristics such as Speed of response, Fidelity, Lag and Dynamic error.	L	CO1	Understand	1
Types of Errors	Compare different types of errors – Gross error, Systematic error and Random error.	L	CO1	Understand	2
Moving Coil and Moving Iron instruments	Explain the working of moving coil instruments and moving iron instruments.	L	CO1	Understand	2
PMMC Galvanometer	Describe the construction and working of PMMC galvanometer	L	CO1	Understand	2
Module II					

Types of Bridges	List DC and AC bridges	L	CO2	Remember	1
DC Bridge	Describe the working of Wheatstone's bridge with circuit diagram.	L	CO2	Understand	2
AC Bridges	Explain the working of Maxwell's bridge, Hay's bridge, Schering bridge and Wien bridge using circuit diagrams.	L	CO2	Understand	3
Display devices	Describe the working of CRO and DSO with block diagram.	L	CO2	Understand	3
Display devices	Explain the working of LED and LCD displays.	L	CO2	Understand	3
Module III					
Transducer and classification	Define a Transducer	L	CO3	Understand	1
Transducer and classification	Classify the transducers on the basis of transduction form used, passive and active, analog and digital, primary and secondary, mechanical and electrical (definitions and examples only)	L	CO3	Understand	2
Resistive transducers	Principle of operation.	L	CO3	Understand	1
Linear Potentiometer	Describe the construction and working of linear potentiometer.	L	CO3	Understand	1
Linear Potentiometer	Describe the loading effect of potentiometer	L	CO3	Understand	1
Linear Potentiometer	Simple problems on linear potentiometer circuit to calculate output voltage, current, and resistance value	L	CO3	Understand	2
Flex transducer	Explain the working of Flex transducer	L	CO3	Understand	1
LDR	Describe the construction and working of LDR	L	CO3	Understand	1
Strain Gauge	Describe the principle of operation of Strain gauge	L	CO3	Understand	1
Strain Gauge	Compare different types of Strain gauges - bonded, unbonded and semiconductor types.	L	CO3	Understand	2
Strain Gauge	Describe Weight measurement using strain gauge bridge circuit.	L	CO3	Understand	2
Module IV					
LVDT	Describe the construction and working of LVDT.	L	CO4	Understand	2
Eddy current transducer	Explain the principle of operation of eddy current transducer	L	CO4	Understand	2
Capacitive transducer	Describe the principle of operation of variable area, variable distance and variable dielectric capacitive transducers.	L	CO4	Understand	2
Capacitive transducer	Explain the level measurement using Capacitive transducer.	L	CO4	Understand	1
Piezoelectric transducer	Describe the principle of operation of Piezoelectric transducer and its equivalent circuit.	L	CO4	Understand	2
Hall effect transducer	Explain the principle of operation of Hall effect transducers.	L	CO4	Understand	2
Ultrasonic transducer	Describe the principle of operation of an ultrasonic transducer.	L	CO4	Understand	1
Ultrasonic transducer	Explain level measurement using Ultrasonic transducer	L	CO4	Understand	1
Photoelectric transducers	List different types of photoelectric transducers	L	CO4	Remember	1
Photoelectric transducers	Explain Photoelectric effect	L	CO4	Understand	1
Photoelectric transducers	Explain the principle of operation and working of photodiodes, phototransistors and solar cells.	L	CO4	Understand	2

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Read basic definitions of measurement, instrumentation, and instrument from the textbook. Prepare a one-page note explaining measurement with two real-life examples.	6

2	Compare direct & indirect and absolute & relative methods of measurement. Draw a neat block diagram of a basic measurement system and explain each block in own words.	6
3	Prepare a comparison table for: • Analog vs Digital instruments • Mechanical vs Electrical vs electronic instruments Identify at least 5 instruments used in daily life and classify them.	6
4	Study static and dynamic characteristics of instruments. Solve simple numerical problems on accuracy, error, and sensitivity. Watch one video lecture on measurement errors and write a short summary.	6
Module II		
5	Draw the circuit diagram of Wheatstone's bridge and write its working principle. List practical applications of Wheatston's bridge in instrumentation.	6
6	Study AC bridges (Maxwell's, Hay's, Schering, Wien). Prepare a summary table showing: • Name of bridge • Quantity measured • Application	6
7	Study CRO and DSO block diagrams. Compare CRO and DSO in terms of features and applications.	6
8	Study LED and LCD displays. Draw the structure of the LED display. Compare LED and LCD display with applications. List applications of displays .	6
Module III		
9	Read the definition and classification of transducers. Prepare a classification chart with examples for each type. Identify some transducers in real life.	6
10	Study the principle of resistive transducers. Draw and explain the construction of a linear potentiometer.	6
11	Solve simple numerical problems on potentiometer: • Output voltage • Resistance calculation Write short notes on loading effect of potentiometer.	6
12	Study LDR and Strain Gauge. Find out different applications of LDR. Study automatic street light system using LDR. Draw the strain gauge bridge circuit used for weight measurement.	6
Module IV		
13	Study LVDT construction and working. Write a short note on pressure measurement using LVDT with diagram.	6
14	Study capacitive transducers and their types. Draw diagrams for: • Variable area • Variable distance • Variable dielectric Write steps for level measurement using capacitive transducer.	6
15	Study piezoelectric, Hall effect, ultrasonic and photoelectric transducers. Prepare short notes on: • Principle • One application for each transducer Watch a video on ultrasonic level measurement and summarize learning.	6
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	A Course in Electrical and Electronic Measurements and Instrumentation.	AK Sawhney	Dhanpath Rai & Sons.
2	Electronic Instrumentation	Kalsi	TMH
3	Electrical Technology	B L Thereja	S Chand
4	Transducers and Instrumentation	D.V.S. Murthy	PHI. Pvt. Ltd.

5	Sensors and Transducers	D. Patranabis	PHI. Pvt. Ltd.
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REFERENCE			
Sl. No.	Title	Author	Publication
1	Modern Electronic Instrumentation and Measurement Techniques	William D Cooper	TMH
2	Electronic Instrumentation and Measurements	David A. Bell	Second Edition, Oxford University Press.
3	Transducer engineering	Ranganathan S	Allied publishers.
4	Principles of Industrial Instrumentation	D. Patranabis	Tata McGraw-Hill. Education,
5	Instrumentation, Measurement and Analysis	B. C. Nakra, K. K. Chaudhry	Tata McGraw-Hill Publishers.
6	Mechanical and Industrial Measurements	R.K. Jain	Khanna Publishers.

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	www.ieee-ims.org	I & II
2	www.nptel.ac.in/courses	I & II
3	https://www.electronicweekly.com/	I & II
4	http://www.instrumentationtoolbox.com/	I & II
5	https://www.slideshare.net/manash234/classification-of-transducers	III
6	https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf	III
7	http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html	IV
8	http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt-tutorial.pdf	IV
9	https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html	IV
10	https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers-work/	IV
11	https://www.ctscorp.com/resource-center/tutorials/piezo-basics/	IV

Assessment Methodology - Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50	50	60
Converted to						20	20	
Marks	5	15				20		60
Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		Total Marks
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60

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